Formal and stereotypical gender in gender agreement: evidence for masculine generic

In gendered languages like Italian nouns' gender reflect a formal (lexical) feature of (most) nouns; in the case of role nouns, it might also reflect semantic gender, i.e., a transparent relationship between the referent's gender and the gender of nouns/pronouns. From a sociolinguistic perspective, role nouns also involve a gender bias, i.e., stereotypical expectations (traditionally, *doctor* is male-biased, *nurse* is female-biased). All these factors might play a role in the process of grammatical agreement, i.e., the coordinate gender assignment across elements in the sentence. Previous studies show that grammatical gender takes precedence over semantic agreement in anaphoric dependencies in Italian [1]. Other studies also suggest that readers integrate (gender) stereotypes, experiencing a clash when the morphological gender and the stereotypical gender associated with a role noun do not match [2]. We aim to contribute to this debate by testing how gendered forms are processed in subject-verb agreement in reading, to answer these research questions: (Q1) Are masculine and feminine forms equally involved in the process of morphological subject-verb agreement? (Q2) Are masculine and feminine forms equally permeable to gender stereotypes associated with nouns referring to people? **Our study.** We conducted two eye-tracking reading studies in Italian. Study 1 (N=54) tested 90 inanimate nouns, cf. (1); Study 2 (N=53) 133 role nouns (83 bigender, *artista*, artist; 50 gendered, *educatore*/*educatrice,* educator, cf. (2). Each noun was followed by a gender-marked verb (past participle), either in masculine or feminine form (*arrivato*/*arrivata*, arrived), rotated in a Latin-square design. Statistical analyses of reading time measures were carried out using mixed-effects linear/logistic regression models. To address Q1, we first compared Study 1 and 2. We show: (i) a symmetric pattern of agreement mismatch between +M and +F forms when the noun is inanimate; (ii) an asymmetric pattern for gendered nouns referring to people (Fig. 1). Specifically, the cost of reading a +M verb after a +F role noun is much reduced in total time measures (compared to the cost of a +F verb after a +M role noun), and it becomes nonexistent or reversed in go past and first pass time measures. This is not the case: (i) when the noun is inanimate; (ii) when the verb is +F: the cost of reading a +F verb after a +M noun remains high, and similar to the mismatch effect found for inanimate nouns. As for Q2, we modeled reading measures in bigender nouns including a previous norming rate as predictor (Authors, 2025). A consistent reduced cost for +M verbs (vs. +F) was found across measures. Late measures of processing also revealed a significant interaction of Gender Bias*Verb Gender (Fig. 2): male-biased and unbiased nouns show a cost if followed by a +F verb, which decreases for highly female-biased nouns (e.g., *badante*); the effect of noun's gender bias is much reduced for +M verbs, and more constant across nouns. This effect was further corroborated in an Acceptability Judgment Study on the same materials (Fig. 3). **Conclusions.** We show that the cost of gender mismatch is greater when a +F verb follows a +M noun than vice versa, but this only happens in the case of nouns referring to people. This effect cannot be reduced to frequency and does not emerge when gender is a formal property of the noun (as for inanimate nouns). When referring to people, the +M form can be considered ambiguous between a gender-specific and a gender-generic or underspecified form. +F forms, instead, entail femaleness [3] and can oonly be specific. If the noun is inanimate, +M and +F are formal traits of the nouns, thus gender is always specific and requires formal agreement. In the case of bigender nouns, we show another asymmetry: +F forms are more sensitive to stereotypical expectations than +M: RTs are longer when a +F verb follows a bigender noun that is perceived as referred to a man, and it is reduced when a +F verb follows a female-biased bigender noun. Our explanation is that readers engage in gender processing when the noun or the verb is marked (i.e., +F if referring to people, +M/+F for singular inanimate nouns), but gender might remain underspecified (i.e., gender generic) in the case of +M forms referring to people. The implications of our findings will be discussed (i) in the context of the ongoing debate on inclusive language and (ii) with respect to prior evidence against generic masculine [4].

Example sentences in Study 1 (inanimate nouns) and Studies 2-3 (role nouns).

(1) a. Durante la prova, l'antidoto$_{[+M]}$ si è rovesciato$_{[+M]}$/rovesciata$_{[+F]}$ sul pavimento.
During the test, the antidote spilled on the floor.

b. Dopo l'incidente, l'amnesia$_{[+F]}$ si è manifestato$_{[+M]}$/manifestata$_{[+F]}$ subito.
After the accident, the amnesia manifested (itself) immediately.

(2) Ieri l'artista$_{[M/F]}$ / educatore$_{[+M]}$ / educatrice$_{[+F]}$ è arrivato$_{[+M]}$/arrivata$_{[+F]}$ tardi.
Yesterday, the artist/educator arrived late.

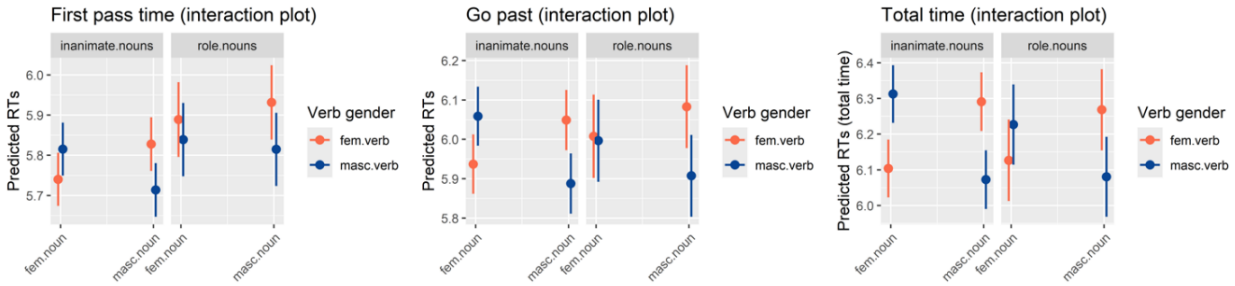

Figure 1. Interaction plots of *lmer* on reading measures of the gendered verb depending on noun's type (inanimate vs. gendered role nouns) and noun's gender. Formula: lmer(log(RT) ~ length.c+zipf.c+ trial.ord+(noun.type*noun.gender*verb.gender) + (1|sub) + (1|item). In all models the three-way interaction was significant (FP: Est.=-0.12, *SE*=0.04, *t*=-2.92, *p*=.003; GP: Est.=-0.12, *SE*=0.05, *t*=-2.24, *p*=.025; TT: Est.=-0.14, *SE*=0.05, *t*=-2.79, *p*=.005).

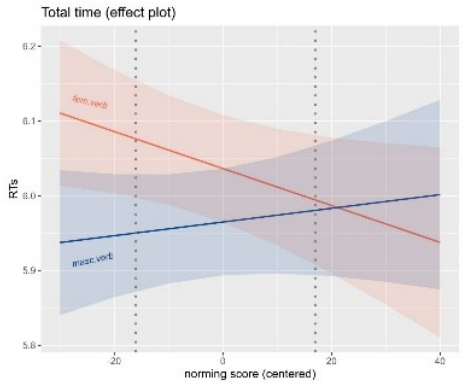

Figure 2. Interaction plot of Verb*Gender Bias for bigendered role nouns (Study 2).

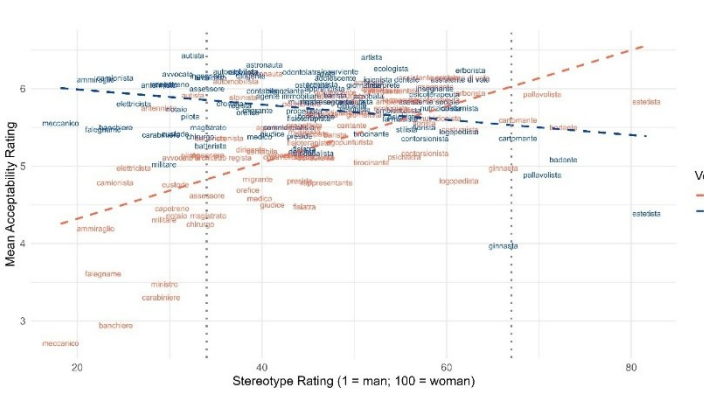

Figure 3. Interaction plot of Verb*Gender Bias for bigendered role nouns (blue=M) in an Acceptability Judgment Study (Study 3).

References [1] Cacciari, C., Carreiras, M., & Cionini, C.B. (1997). When words have two genders: Anaphor resolution for Italian functionally ambiguous words. Journal of memory and language, 37(4), 517-532; [2] Carreiras, M., Garnham, A., Oakhill, J., & Cain, K. (1996). The use of stereotypical gender information in constructing a mental model: Evidence from English and Spanish. The Quarterly Journal of Experimental Psychology (Section A), 49,639-663; [3] Percus, O. (2011). Gender features and interpretation: a case study. Morphology, 21(2), 167-196; [4] Gygax, P., Gabriel, U., Sarrasin, O., Oakhill, J., & Garnham, A. (2008). Generically intended, but specifically interpreted: When beauticians, musicians, and mechanics are all men. Language and cognitive processes, 23(3), 464-485.